

A Möbius–Descartes Proof of Unique Root Location for an Eighth-Degree Stationary Polynomial

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Abstract

For every $0 < q < 1$ and $s, t > 0$, we prove that the eighth-degree polynomial arising in a numerical-radius problem for a periodic weighted shift has exactly one real zero in $(1/3, 1)$; moreover, this zero is simple. This answers Open Question 1 of Ghosh, Birbonshi, and Ojha [arXiv:2608.17486v1]. A fractional-linear change of variables maps the target interval onto the positive half-line. The nonzero coefficients of the transformed polynomial have exactly one sign variation: an adjacent coefficient comparison reduces the needed sign nesting to five positive determinants, certified by elementary inequalities, and five positive polynomials, certified by their Bernstein coefficients. Descartes' rule of signs then gives the result, including simplicity.

1 Statement and context

Let $0 < q < 1$ and $s, t > 0$, and set

$$\begin{aligned} g_{q,s,t}(z) = & tq^7z^8 + q^6(tq - 1)z^7 + 3q^5(q - s)z^6 + 5q^4(sq - 1)z^5 \\ & + q^3(7q - 9t)z^4 + 7q^2(tq - 1)z^3 + 5q(q - s)z^2 + 3(sq - 1)z + 1. \end{aligned}$$

This polynomial occurs as the stationary-point polynomial for a lower bound on the numerical radius of a class of weighted shifts [1]. Theorem 2.5 of that paper proves uniqueness in $(0, 1)$ under the additional restrictions $q < s, t < 1/q$, while Theorem 2.12 together with the displayed value at $z = 1/3$ excludes zeros in $(0, 1/3]$ for all admissible parameters. Its Open Question 1 asks whether the zero in $(1/3, 1)$ is unique without the additional restrictions.

Fractional-linear normalizations also arise elsewhere in operator-theoretic numerical-range questions; for example, [2] uses a Möbius map to identify a convex-lens spectral-set problem with a sectorial numerical-range problem. That work concerns a different operator inequality. Here the transformation is used only for real-root localization, and the argument after the transformation is algebraic.

Theorem 1. *For every $0 < q < 1$ and $s, t > 0$, the polynomial $g_{q,s,t}$ has exactly one real zero in $(1/3, 1)$. This zero has algebraic multiplicity one.*

2 Coefficient sign nesting

Use the mutually inverse changes of variables

$$z = \frac{x + 1}{x + 3}, \quad x = \frac{3z - 1}{1 - z}.$$

They give an increasing bijection from $x > 0$ onto $1/3 < z < 1$. Define

$$H(x) = (x+3)^8 g_{q,s,t} \left(\frac{x+1}{x+3} \right) = \sum_{i=0}^8 c_i x^i.$$

Direct expansion at the two ends of the coefficient sequence gives

$$c_0 = 4q(tq^6 + 6q^5 + 27sq^4 + 108q^3 + 243tq^2 + 486q + 729s) > 0, \quad (1)$$

$$c_7 = 2(q-1)(q+1)(q^2+1)(9q^3t + 13q^2 + 17qs + 21) < 0, \quad (2)$$

$$c_8 = 2(q-1)(q+1)(q^2+1)(q^3t + q^2 + qs + 1) < 0. \quad (3)$$

For $1 \leq i \leq 6$, write

$$c_i = 2A_i, \quad A_i = T_i t + S_i s + R_i.$$

Putting $u = q^2$, the three coefficient families are given by

$$T_i = q^3 \tau_i(u), \quad S_i = q \sigma_i(u), \quad R_i = \rho_i(u),$$

where the following table is an exact expansion of H .

i	$\tau_i(u)$	$\sigma_i(u)$	$\rho_i(u)$
1	$15u^2 + 2025$	$315u^2 + 3645$	$79u^3 + 1107u^2 + 3321u - 2187$
2	$49u^2 + 3375$	$777u^2 + 3159$	$223u^3 + 2367u^2 + 4401u - 5103$
3	$91u^2 + 2853$	$1051u^2 + 837$	$351u^3 + 2735u^2 + 2817u - 5103$
4	$105u^2 + 1263$	$845u^2 - 405$	$335u^3 + 1851u^2 + 825u - 2835$
5	$77u^2 + 251$	$409u^2 - 369$	$197u^3 + 745u^2 + 19u - 945$
6	$35u^2 - 3$	$115u^2 - 115$	$69u^3 + 173u^2 - 53u - 189$

For $1 \leq i \leq 5$, introduce the adjacent determinants

$$\Delta_i = S_i T_{i+1} - T_i S_{i+1}, \quad N_i = R_i T_{i+1} - T_i R_{i+1}.$$

Expansion and factorization give

$$\begin{array}{c|l} i & \Delta_i \\ \hline 1 & 540q^4(9 - q^2)(q^2 + 9)(135 - 7q^4), \\ 2 & 8q^4(2401q^8 - 135486q^4 + 773469), \\ 3 & 4q^4(8365q^8 - 239658q^4 + 553149), \\ 4 & 56q^4(395q^8 - 5302q^4 + 6507), \\ 5 & 4q^4(1365q^8 - 8538q^4 + 7493). \end{array} \quad (4)$$

Each Δ_i is positive. For the last four rows, set $v = q^4$. The polynomial in parentheses is decreasing on $0 \leq v \leq 1$ and its value at $v = 1$ is, respectively,

$$640384, \quad 321856, \quad 1600, \quad 320.$$

The first row is positive directly, since $0 < q < 1$.

The positivity of the N_i has a similarly explicit certificate. Let

$$B_{k,5}(u) = \binom{5}{k} u^k (1-u)^{5-k} \quad (0 \leq k \leq 5)$$

be the degree-five Bernstein basis. Exact conversion from the power basis gives

$$N_i = 2q^3 \sum_{k=0}^5 b_{ik} B_{k,5}(u), \quad u = q^2, \quad (5)$$

with coefficient rows

i	b_{i0}	b_{i1}	b_{i2}	b_{i3}	b_{i4}	b_{i5}
1	1476225	1705860	9405558/5	1997568	10263456/5	2046080
2	1331883	8183754/5	9035118/5	1828404	8473664/5	1409536
3	821583	4709988/5	4785954/5	852576	625440	288128
4	240975	1296414/5	1239102/5	201428	121376	17920
5	25137	132308/5	122886/5	94128/5	9824	256

Every displayed number is positive, and the Bernstein basis functions are nonnegative on $[0, 1]$ and sum to one. Hence

$$\Delta_i > 0 \quad \text{and} \quad N_i > 0 \quad (1 \leq i \leq 5). \quad (6)$$

Lemma 2. For $1 \leq i \leq 5$,

$$A_i \leq 0 \implies A_{i+1} < 0.$$

Proof. The table shows that $T_1, \dots, T_5$ are positive. For $1 \leq i \leq 4$, cancellation of the t -term gives

$$A_i - \frac{T_i}{T_{i+1}} A_{i+1} = \frac{\Delta_i s + N_i}{T_{i+1}} > 0. \quad (7)$$

Thus $A_i \leq 0$ forces $A_{i+1} < 0$. The same argument applies for $i = 5$ when $T_6 > 0$. If $T_6 \leq 0$, then

$$S_6 = 115q(q-1)(q+1)(q^2+1) < 0$$

and

$$R_6 = (q-1)(q+1)(3q^2+7)(23q^2+27) < 0.$$

Since $s, t > 0$, these signs give $A_6 = T_6 t + S_6 s + R_6 < 0$ directly. This covers the remaining case, including $T_6 = 0$. $\square$

3 Proof of the theorem

Proof of Theorem 1. By (1), the coefficient sequence of H starts positive, and by (2)–(3) it ends negative. Lemma 2 says that once one of the middle coefficients $c_1, \dots, c_6$ is nonpositive, every subsequent middle coefficient is strictly negative. Consequently, after zero coefficients are omitted, the sequence $(c_0, c_1, \dots, c_8)$ has exactly one sign variation.

Descartes' rule of signs now implies that H has exactly one positive real zero counted with algebraic multiplicity. In particular, that zero is simple. Finally, $x \mapsto (x+1)/(x+3)$ is a bijection from $(0, \infty)$ to $(1/3, 1)$, its derivative is nonzero there, and the clearing factor $(x+3)^8$ never vanishes there. Zeros and their multiplicities are therefore preserved under the transformation. The unique positive zero of H corresponds to one simple zero of $g_{q,s,t}$ in $(1/3, 1)$. $\square$

Computational and AI assistance

Exact symbolic computation in Python 3.13 with SymPy 1.14 was used during discovery and checking to expand H , verify the identities (1)–(5), and independently test the positivity certificates with Sturm chains. Exact rational parameter grids were also searched for counterexamples, but no finite search is used in the proof. An OpenAI Codex language model assisted with proof exploration, verifier code, exact-identity checking, and manuscript preparation. The displayed identities and inequalities constitute the proof; software success is not treated as a substitute for them.

References

- [1] Arobinda Ghosh, Riddhick Birbonshi, and Sarita Ojha. On the numerical radius of a class of weighted shift operators. *arXiv preprint arXiv:2608.17486*, 2026. Version 1, 18 August 2026; <https://arxiv.org/abs/2608.17486>.
- [2] Zijian Zeng. Fixed convex-lens spectral constants: Möbius reduction, sharp model theorems, and angle-dependent bounds. *Preprints*, August 2026. doi:10.20944/preprints202608.1272.v1.